

FLEET RULE OF THE ROAD TEST Feb 2015

This test is based on: The International Regulations for Preventing Collisions at Sea (1972) with amendments in force up to December 2009 as detailed in "A Seaman's Guide Pocket Book of The International Regulations for Preventing Collisions at Sea" (BR 453 Supp), ATP1(E) Vol I and the Mariners Handbook (NP100).

BR454 "A Guide to the Collision Avoidance Rules" 6th Edition by Cockcroft and Lameijer was categorised as obsolescent by NCHQ in August 2011. Care should be taken when utilising the 6th Edition as the information contained within may not reflect the most recent changes to the regulations.

The test is designed to be given to any Warfare Officer, Deck officer, Officer undertaking bridge watch-keeping duties or Young Officer/Cadet. The test should be administered under written exam conditions, the results should be recorded in the ROR/training logbook for inspection by FLEET/FOST navigation staff.

In this test, any change in the status of the ship or of the environment will be indicated in the script. .

Rules 1 to 19, 34, 35 and Annex IV must be known instinctively for instant application. A thorough knowledge of the other rules is required. The pass mark is 90% (50 Marks).

Answers are provided for uniformity of marking. This test is completely scenario based.

You are the SSD Officer of the Watch of a T23 frigate (length 133 meters), having just passed Outer Spit Buoy and entered the Solent to make a transit to the Needles Channel. It is the forenoon watch, the weather is fine with good visibility, however a cold front is forecast. Shipping is moderate.

1. As you alter course to the west the hovercraft from Southsea is on a converging course just ahead of your starboard beam. You are 25 degrees on her PORT bow. What action is required by the Rules if a risk of collision develops? **(R. 15) (1)**

2. As you pass south of Ryde Middle you sight a small, slow moving tug and partially submerged tow in the channel ahead, length of tow >200m. You wish to overtake on the port side. (He will need to manoeuvre to allow you to pass.)

a. As you are unable to raise him on VHF, what sound signal can be made to signal your intention? **(R. 34c) (2)**

b. If the master of the vessel is in agreement, how can this be indicated by sound signal? **(R. 34c)** (1)

c. What shapes would you expect to see in the rigging of the tug and tow? **(R. 24)** (3)

3. As you pass Cowes you see the Tall Ships' race further down the Solent inbound to Southampton, including both square-rigged and fore and aft rigged vessels. Your Captain asks you how you would determine the windward side of either type of vessel? **(R. 12b)** (2)

4. There are also numerous small pleasure craft on the water, obviously enjoying the spectacle of the Tall Ships' race. The Captain draws your attention to several sailing vessels displaying shapes in the rigging consisting of a conical shape, apex down. She asks what this means? **(R. 25e)** (1)

5. Continuing on your passage you sight a large merchant vessel at R10, you are fine on his port bow and he appears to be transiting along the centreline of the fairway.

a. How should a vessel navigate within a narrow channel or fairway? **(R. 9a)** (1)

b. You hear a conversation between the large merchant vessel and a small pleasure craft approaching the limits of the buoyed channel. The merchant vessel states that she is CBD. What shapes by day, and lights by night would you expect the merchant vessel to display? **(R. 28)** (2)

c. What do the rules state concerning the actions of the small pleasure craft approaching the buoyed channel? **(R. 9b)** (1)

d. What do the rules state regarding a vessel wishing to cross the main channel in such a situation, and what could the merchant vessel do if in doubt about the intention of the crossing vessel? **(R. 9d, 34d)** (2)

6. As you pass the merchant vessel you notice a small vessel crossing his stern displaying a red and white vertical striped flag (flag hotel).

a. What does this indicate? (1)

b. What lights would you expect this vessel to exhibit at night, when underway? **(R. 29)** (2)

7. You continue down past the Needles without further incident. On falling out from SSD you take the remainder of the watch just as the rain starts and visibility begins to reduce. You call the CO to inform her and commence your check-offs for entering restricted visibility.

a. What sound signals will you commence? **(R. 35a)** (1)

b. You switch on the appropriate navigation lights but on checking the port sidelight it appears dim. What are the visibility ranges and horizontal sectors for the lights applicable to your vessel? **(R. 21, 22)** (3)

- c. What do the rules state about safe speed in restricted visibility?
(R. 19b) (2)

8. After discussion with the Navigation Officer you inform the CO that you intend to continue down the nav track at 12 knots. Shortly afterwards you detect a contact on radar at G10 range 5 miles. The vessel's course is parallel to your own but the CPA is on your starboard beam at 9 cables.

- a. What do the rules state about this situation and the actions you should take? **(R. 19d) (2)**

- b. You take the appropriate action and continue with your watch. Soon afterwards you hear a fog signal consisting of one prolonged followed by two short blasts. What types of vessel could this be? **(R. 35c) (3)**

- c. On checking the radar you see that the previously reported contact has altered course and is now at G30 range 1.5 miles and is on a steady bearing. What action shall you take in accordance with the rules? **(R. 19e) (3)**

- d. The contact closes to 1 mile and you sight the following lights:

W
G
W
G

What type of vessel is this and in what activity is it engaged? **R26(b)(2)**

9. The vessel alters course again and you pass clear, proceeding along the nav track. The cold front passes through and visibility increases to 12nm so you revert back to cruising watches. Shortly you spot a small merchant vessel at red 25, range 5 nm, you assess that you are 60 degrees on her starboard bow and her bearing is steady.

a. In accordance with the Rules what do you expect the vessel to do and how shall she act? **(R. 15, 16)** (2)

b. What should you do? **(R. 17a)** (1)

c. The vessel is now at 2 nm and has not altered course. You are unsure of his intentions, what should you do? **(R. 34d)** (1)

d. When may you to take action? **(R. 17a)** (1)

e. When must you take action? **(R. 17b)** (1)

10. At 1.5 nm vessel alters course and passes clear. The end of your watch is rapidly approaching but you note that the Ship is due to enter a TSS shortly. While preparing for the handover you discuss Rule 10 with the 2OOW.

a. How shall a vessel navigate near the terminations of a TSS? **(R. 10f)** (1)

b. How shall a vessel using a TSS navigate? **(R. 10b)** (2)

c. Which types of vessel are to avoid impeding the passage of vessels following a traffic lane? **(R. 10i,j)** (3)

d. Which types of vessel are exempt from complying with this rule, and to what extent? **(R. 10k,l)** (2)

11. As a diligent OOW you take your instructional duties very seriously and decide to set the 2OOW some homework prior to your next watch together. You ask him to find 6 signals that could be used to indicate distress or need of assistance. Of course, you already know the answer. **(Annex VI)** (6)

You hand over the watch and finally head down to your cabin for some rest before another busy day commences.

Total 55 marks

ANSWERS

1. As you alter course to the west the hovercraft from Southsea is on a converging course just ahead of your starboard beam. You are 25 degrees on her PORT bow. What action is required by the Rules if a risk of collision develops? **(R. 15) (1)**

When two power driven vessels are crossing so as to involve risk of collision, the vessel which has the other on her own starboard side shall keep out of the way and shall, if the circumstances of the case admit, avoid crossing ahead of the other vessel (1).

2. As you pass south of Ryde Middle you sight a small, slow moving tug and partially submerged tow in the channel ahead, length of tow >200m. You wish to overtake on the port side. (He will need to manoeuvre to allow you to pass.)

- a. As you are unable to raise him on VHF, what sound signal can be made to signal your intention? **(R. 34c) (2)**

A vessel intending to overtake another shall in compliance with Rule 9(e)(i) indicate her intention by the following signals on her whistle:

- **two prolonged blasts followed by two short blasts (1) to mean 'I intend to overtake you on your port side' (1).**

- b. If the master of the vessel is in agreement, how can this be indicated by sound signal? **(R. 34c) (1)**

The vessel about to be overtaken when acting in accordance with Rule 9(e)(i) shall indicate her agreement by the following signal on her whistle:

- **one prolonged, one short, one prolonged and one short blast, in that order (1).**

- c. What shapes would you expect to see in the rigging of the tug and tow? **(R. 24) (3)**

A power driven vessel when towing shall exhibit:

- **when the length of the tow exceeds 200m (1), a diamond shape where it can best be seen.**

An inconspicuous, partly submerged vessel or object, or combination of such vessels or objects being towed, shall exhibit:

- **a diamond shape at or near the aftermost extremity of the last vessel or object being towed (1) and if the length of the tow exceeds 200m an additional diamond shape where it can best be seen and located as far forward as practicable (1).**

3. As you pass Cowes you see the Tall Ships' race further down the Solent inbound to Southampton, including both square-rigged and fore and aft rigged vessels. Your Captain asks you how you would determine the windward side of either type of vessel? **(R. 12b) (2)**

For the purpose of this Rule the windward side shall be deemed to be the side opposite to that on which the mainsail is carried (1) or, in the case of a square-rigged vessel, the side opposite to that on which the largest fore-and-aft sail is carried (1).

4. There are also numerous small pleasure craft on the water, obviously enjoying the spectacle of the Tall Ships' race. The Captain draws your attention to several sailing vessels displaying shapes in the rigging consisting of a conical shape, apex down. She asks what this means? **(R. 25e) (1)**

A vessel proceeding under sail when also being propelled by machinery shall exhibit forward where it can best be seen a conical shape, apex downwards.

5. Continuing on your passage you sight a large merchant vessel at R10, you are fine on his port bow and he appears to be transiting along the centreline of the fairway.

a. How should a vessel navigate within a narrow channel or fairway? **(R. 9a) (1)**

A vessel proceeding along the course of a narrow channel or fairway shall keep as near to the outer limit of the channel or fairway which lies on her starboard side as is safe and practicable (1).

b. You hear a conversation between the large merchant vessel and a small pleasure craft approaching the limits of the buoyed channel. The merchant vessel states that she is CBD. What shapes by day, and lights by night would you expect the merchant vessel to display? **(R. 28) (2)**

In addition to the lights prescribed for a power-driven vessel [of her length] may exhibit where they can best be seen three all round red lights in a vertical line (1) or a cylinder (1).

c. What do the rules state concerning the actions of the small pleasure craft approaching the buoyed channel? **(R. 9b) (1)**

A vessel of less than 20m in length or a sailing vessel shall not impede the passage of a vessel which can safely navigate only within a narrow channel or fairway (1).

d. What do the rules state regarding a vessel wishing to cross the main channel in such a situation, and what could the merchant vessel do if in doubt about the intention of the crossing vessel? **(R. 9d, 34d) (2)**

A vessel shall not cross a narrow channel or fairway if such crossing impedes the passage of a vessel which can safely navigate only within such channel or fairway (1). The latter vessel may use the sound signal prescribed in Rule 34(d) if in doubt as to the intention of the crossing vessel.

...the vessel in doubt shall immediately indicate such doubt by fiving at least five short and rapid blasts on the whistle (1). Such signal may be supplemented by a light signal of at least five short and rapid flashes.

6. As you pass the merchant vessel you notice a small vessel crossing his stern displaying a red and white vertical striped flag (flag hotel).

a. What does this indicate? (1)

Vessel engaged on pilotage duties (1).

b. What lights would you expect this vessel to exhibit at night, when underway? **(R. 29) (2)**

A vessel engaged on pilotage duty shall exhibit:

- **at or near the masthead, two all round lights in a vertical line, the upper being white and the lower red (1);**
- **when underway, in addition, sidelights and a sternlight (1).**

7. You continue down past the Needles without further incident. On falling out from SSD you take the remainder of the watch just as the rain starts and visibility begins to reduce. You call the CO to inform her and commence your check-offs for entering restricted visibility.

a. What sound signals will you commence? **(R. 35a) (1)**

A power-driven vessel making way through the water shall sound at intervals of not more than 2 minutes (1/2) one prolonged blast (1/2).

b. You switch on the appropriate navigation lights but on checking the port sidelight it appears dim. What are the visibility ranges and horizontal sectors for the lights applicable to your vessel? **(R. 21, 22) (3)**

- **Masthead light...unbroken light from right ahead to 22.5 degrees abaft the beam on either side of the vessel (1/2). Minimum range 6nm (1/2).**
- **Sidelights...unbroken light from right ahead to 22.5 degrees abaft the beam on its respective side (1/2). Minimum range 3nm (1/2).**
- **Sternlight...unbroken light from 67.5 degrees from right aft on each side of the vessel (1/2). Minimum range 3nm (1/2).**

c. What do the rules state about safe speed in restricted visibility? **(R. 19b) (2)**

Every vessel shall proceed at a safe speed adapted to the prevailing circumstances and conditions of restricted visibility (1). A power-driven vessel shall have her engines ready for immediate manoeuvre (1).

8. After discussion with the Navigation Officer you inform the CO that you intend to continue down the nav track at 12 knots. Shortly afterwards you detect a contact on radar at G10 range 5 miles. The vessel's course is parallel to your own but the CPA is on your starboard beam at 9 cables.

a. What do the rules state about this situation and the actions you should take? **(R. 19d)** (2)

A vessel which detects by radar alone the presence of another vessel shall determine if a close-quarters situation is developing and/or risk of collision exists. If so, she shall take avoiding action in ample time, provided that when such action consists of an alteration of course, so far as possible the following shall be avoided:

- **an alteration of course to port for a vessel forward of the beam other than for a vessel being overtaken (1);**
- **an alteration of course towards a vessel abeam or abaft the beam (1).**

b. You take the appropriate action and continue with your watch. Soon afterwards you hear a fog signal consisting of one prolonged followed by two short blasts. What types of vessel could this be? **(R. 35c)** (3)

A vessel not under command (1/2), a vessel restricted in her ability to manoeuvre (1/2), a vessel constrained by her draught (1/2), a sailing vessel (1/2), a vessel engaged in fishing (1/2) or a vessel engaged in towing or pushing another vessel(1/2).

c. On checking the radar you see that the previously reported contact has altered course and is now at G30 range 1.5 miles and is on a steady bearing. What action shall you take in accordance with the rules? **(R. 19e)** (3)

Except where it has been determined that a risk of collision does not exist, every vessel which hears apparently forward of her beam the fog signal of another vessel, or which cannot avoid a close quarters situation with another vessel forward of her beam (1), shall reduce her speed to the minimum at which she can be kept on her course (1), She shall if necessary take all her way off and in any event navigate with extreme caution until danger of collision is over (1).

d. The contact closes to 1 mile and you sight the following lights:

W
G
W
G

What type of vessel is this and in what activity is it engaged? **(R.26(b))** (2)

Vessel engaged in trawling (1/2), underway (1/2), seen from the STBD side (1/2), probably more than 50m in length (1/2).

9. The vessel alters course again and you pass clear, proceeding along the nav track. The cold front passes through and visibility increases to 12nm so you revert back to cruising watches. Shortly you spot a small merchant vessel at red 25, range 5 nm, you assess that you are 60 degrees on her starboard bow and her bearing is steady.

- a. In accordance with the Rules what do you expect the vessel to do and how shall she act? **(R. 15, 16) (2)**

When two power-driven vessels are crossing so as to involve risk of collision, the vessel which has the other on her own starboard side shall keep out of the way and shall, if the circumstances of the case admit, avoid crossing ahead of the other vessel (1).

Every vessel which is directed to keep out of the way of another vessel shall, so far as possible, take early and substantial action to keep well clear (1).

- b. What should you do? **(R. 17a) (1)**

Where one of two vessels is to keep out of the way the other shall keep her course and speed (1).

- c. The vessel is now at 2 nm and has not altered course. You are unsure of his intentions, what should you do? **(R. 34d) (1)**

...the vessel in doubt shall immediately indicate such doubt by fiving at least five short and rapid blasts on the whistle. Such signal may be supplemented by a light signal of at least five short and rapid flashes (1).

- d. When may you to take action? **(R. 17a) (1)**

The latter vessel may, however, take action to avoid collision by her manoeuvre alone, as soon as it becomes apparent to her that the vessel required to keep out of the way is not taking appropriate action in compliance with these Rules (1).

- e. When must you take action? **(R. 17b) (1)**

When, from any cause, the vessel required to keep her course and speed finds herself so close that collision cannot be avoided by the action of the give-way vessel alone, she shall take such action as will best aid to avoid collision (1).

10. At 1.5 nm vessel alters course and passes clear. The end of your watch is rapidly approaching but you note that the Ship is due to enter a TSS shortly. While preparing for the handover you discuss Rule 10 with the 2OOW.

- a. How shall a vessel navigate near the terminations of a TSS? **(R. 10f) (1)**

A vessel navigating in areas near the termination of traffic separation schemes shall do so with particular caution (1).

- b. How shall a vessel using a TSS navigate? **(R. 10b) (2)**

A vessel using a traffic separation scheme shall:

- **proceed in the appropriate traffic lane in the general direction of traffic flow for that lane (1/2);**

- so far as practicable keep clear of a traffic separation line or separation zone (1/2);
- normally join or leave a traffic lane at the termination of the lane (1/2), but when joining or leaving from either side shall do so at as small an angle to the general direction of traffic flow as practicable (1/2).

c. Which types of vessel are to avoid impeding the passage of vessels following a traffic lane? (R. 10i,j) (3)

A vessel engaged in fishing (1)....A vessel of less than 20m in length (1) or a sailing vessel (1) shall not impede the (safe) passage of a power-driven vessel following a traffic lane.

d. Which types of vessel are exempt from complying with this rule, and to what extent? (R. 10k,l) (2)

A vessel restricted in her ability to manoeuvre when engaged in an operation for the maintenance of safety of navigation (1/2) in a traffic separation scheme is exempted from complying with this Rule to the extent necessary to carry out the operation (1/2).

A vessel restricted in her ability to manoeuvre when engaged in an operation for the laying, servicing or picking up of a submarine cable (1/2), within a traffic separation scheme, is exempted from complying with this Rule to the extent necessary to carry out the operation (1).

11. As a diligent OOW you take your instructional duties very seriously and decide to set the 2OOW some homework prior to your next watch together. You ask him to find 6 signals that could be used to indicate distress or need of assistance. Of course, you already know the answer. (Annex VI) (6)

Any six of the following:

- a gun or other explosive signal fired at intervals of about a minute;
- a continuous sounding with any fog-signaling apparatus,
- rockets or shells, throwing red stars fired one at a time at short intervals;
- a signal made by radiotelegraphy or by any other signaling method consisting of the group ... _ _ _ ... (SOS) in the Morse Code;
- a signal sent by radiotelegraphy consisting of the spoken word 'mayday';
- the International Code Signal of distress indicated by N.C.;
- a signal consisting of a square flag having above or below it a ball or anything resembling a ball;
- flames on the vessel (as from a burning tar barrel, oil barrel, etc);
- a rocket parachute flare or a hand-flare showing a red light;
- a smoke signal giving off orange-coloured smoke.
- Slowly and repeatedly raising and lowering arms outstretched to each side.
- A distress alert by digital selective calling (DSC) transmitted on;
- VHF70 Or MF/HF on the frequencies 2187.7Khz, 8414.5Khz,4207.5Khz,6312Khz, 12577Khz or 16804.5Khz.
- A ship to shore distress alert transmitted by the ship's Inmarsat or other mobile satellite services provider ship earth station.

- A signal transmitted by emergency position indicating radio beacons
- Approved signals transmitted by radiocommunications systems, including survival craft radar transponders.

You hand over the watch and finally head down to your cabin for some rest before another busy day commences.

Total 55 marks